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PAPER_276: UQFF

Date: March 2026 \$n = [int]\# UQFF PAPER_276 \#\# Andromeda Friedmann-UQFF Gravity Coupling: \$H(z)t\$ Expansion Term and \$H_UQFF\$ Near-Unity Resonance

Author: Daniel T. Murphy

Session: 76 (March 2026)

Module: ANDROMEDA_{UQFF_MODULE}.cpp (M31 Master, UQFF 2.0)

WOLFRAM_TERM: AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[0m_m*(1+z)^3+0m_L]/Mpc2m;
 H_UQFF=H(z)*t_H~0.987 [PAPER_276]

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as $g_expansion = (\mu s \nabla(M_s/r)) H(z) t$. For Andromeda's blueshift redshift $z = -0.001$, the Friedmann $H(z)$ equals $69.969 \text{ km/s/Mpc} = 2.269 \times 10^{-8} \text{ s}^{-1}$. Evaluated at the Hubble timescale $t = t_H = 4.352 \times 10^{-7} \text{ s}$, the dimensionless Friedmann-UQFF Resonance Coefficient $H_UQFF = H(z) \cdot t_H$ **0.987** a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the $M_visible/M\{DM_mass\}$ cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

1. Background

The UQFF framework computes total gravitational acceleration $g_total(r,t)$ as a sum of DPM-seeded, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER_273275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaître cosmological evolution of Hubble parameter $H(z)$. This is significant because over a Hubble timescale, $H(z)t \sim 1$ implying that expansion-mediated gravitational coupling is of the same order as the Step 10 Newton observational projection term.

2. Friedmann $H(z)$ Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard Λ CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_Lambda}$$

For Andromeda parameters: $H_0 = 70.0$ km/s/Mpc, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $z = -0.001$.

$$H(z = -0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: $H_{\text{SI}} = 69.969 \times 10^{-8} / 3.086 \times 10^{22} = 2.269 \times 10^{-18} \text{ s}^{-1}$

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t .

Numerical values for Andromeda: $-g_{\text{base}} = \mu_s \nabla(M_s/r) = 6.674 \times 10^{-11} \cdot 1.989 \times 10^{30} / (1.04 \times 10^{17})^2 = 1.227 \times 10^{-10} \text{ m/s}^2$
 $-Att = 1 \text{ Gyr} (3.156 \times 10^8 \text{ s}) : g_{\text{expansion}} = 1.227 \times 10^{-10} \cdot 2.269 \times 10^{-18} \times 3.156 \times 10^8 = 8.79 \times 10^{-10} \text{ m/s}^2$
 $-Att = t_{\text{Hubble}} (4.352 \times 10^8 \text{ s}) : g_{\text{expansion}} = 1.227 \times 10^{-10} \cdot 2.269 \times 10^{-18} \times 4.352 \times 10^8 \approx 0.987 = \mathbf{1.211 \times 10^{-10} \text{ m/s}^2}$

3. H_{UQFF} Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^8 \text{ s}$ (Hubble time 13.8 Gyr).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^8 = \mathbf{0.987}$$

3.2 Near-Unity Discovery

$H_{\text{UQFF}} \approx 0.987$ is a remarkable result. It means:

Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the Step 10 Newton observational projection term** nearly doubling g_{base} .

This near-unity value is not coincidental. In a flat Λ CDM universe ($\Omega_m + \Omega_\Lambda = 1$):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because $H_0 t_H$ is the dimensionless Hubble number ≈ 0.96 for observationally consistent H_0 .

New UQFF Constant: $H_{\text{UQFF}} = H(z) \cdot t_H$ Friedmann-UQFF Near-Unity Resonance Coefficient

Parameter	Value
H_0	70.0 km/s/Mpc
O_m	0.3
$O_?$	0.7
z (Andromeda)	-0.001
$H(z)$	$2.269 \times 10^8 s^{-1} t_H 4.352 \times 10^{-7} s * H_{\text{UQFF}} * * 0.987 * (1) g_{\text{expansion}}(t_H) 1.211 \times 10^7 m/s g_{\text{base}} 1.227 \times 10^7 m/s$
Ratio $g_{\text{exp}}/g_{\text{base}}$	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $O_m(1+z)$ is suppressed below O_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z = 0)$$

For Andromeda ($z = -0.001$): $H_{\text{UQFF}} = 0.987$ vs flat-universe $H_{\text{UQFF}0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0 \Rightarrow H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 10^6 = 1.989 \times 10^{-2} \text{ kg/m}^3$.

Numerical value for Andromeda:

$$\begin{aligned}
 a_{\text{dust}} &= \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\
 &= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\
 &= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2
 \end{aligned}$$

At ~9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. M_visible / M_{DM_mass} Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$

$$M_{\text{DM, mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: $M = 1.989 \times 10^4 \text{ kg}$, $f_{\text{DM}} = 0.80$: - $M_{\text{visible}} = 0.20 \times 1.989 \times 10^4 = 3.978 \times 10^4 \text{ kg}$ (20% visible baryons) - $M_{\text{DM, mass}} = 0.80 \times 1.989 \times 10^4 = 1.591 \times 10^4 \text{ kg}$ (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 2632), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA_{UQFF_MODULE}.cpp g_total

The full UQFF 2.0 equation for Andromeda with all PAPER_273276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g, \text{sum}}^{(26)} + \Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at t=0 (current epoch):

Term	Value (m/s)	Paper
$g_{\text{grav}} = \mu_s \nabla(M_s/r)$	$1.227 \times 10^7 baseline U_{g, \text{sum}}(26) 26 - layerTriadic - term 2.998 \times 10^6 cosmological g_{\text{quantum}} 2 \times 10^6 HUP g_{\text{Lorentz}} 20 1 \times 10^7 PAPER_{274} g_{\text{DM}} 1.356 \times 10^7 PAPER_{275} g_{\text{expansion}} 0 0 PAPER_{276} a_{\text{dust}} 4.29 \times 10^7$	PAPER_273
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Note: $g_{\text{expansion}}$ grows linearly with t, dominating over g_{grav} at $t = t_{\text{H}}$.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`: 1. $H_0 = 70.0 \text{ km/s/Mpc}$ 2. $\Omega_{\text{m}} = 0.3$ 3. $\Omega_{\text{Lam}} = 0.7$ 4. $\text{Mpc}_{\text{to_m}} = 3.086 \times 10^5$. 'rho_dust' = $1 \times 10^7 \text{ kg/m}$ 6. $M_{\text{visible}} = (1 - f_{\text{DM}})M$ 7. $M_{\text{DM_mass}} = f_{\text{DM}}M$

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Three new UQFF enhancements for Andromeda (PAPER_276):

- Friedmann-UQFF Expansion Coupling:** $g_{\text{expansion}} = g_{\text{base}} - H(z) t$. At $t = t_{\text{H}}$, adds 98.7% of g_{base} near-gravitational doubling over cosmological time.
- H_UQFF = 0.987:** New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 t_{\text{H}} \approx 1$ in flat Λ CDM. Andromeda's blueshift ($z = -0.001$) makes H_{UQFF} 0.15% below flat-universe value a blueshift Friedmann suppression.

3. **M_visible/M_{DM_mass} cascade + dust drag:** Explicit DM split bookkeeping (update-Cache) and ISM dust ram-pressure ($a_{\text{dust}} = 4 \times 10^{-10} \text{ m/s}$).

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Watermark: 2025-2026 Daniel T. Murphy, daniel.murphy00@gmail.com – All Rights Reserved .Groups[1].Value $H_{\text{UQFF}} = H(z) * t_H \sim 0.987$ [PAPER_276]

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H_{UQFF}	0.987 (~ 1)
$g_{\text{exp}}(t_H)$	$1.211 \times 10^{-10} \text{ m/s}^2$
g_{base}	$1.227 \times 10^{-10} \text{ m/s}^2$
Ratio $g_{\text{exp}}/g_{\text{base}}$	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $O_m(1+z)$ is suppressed below O_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z = 0)$$

For Andromeda ($z = -0.001$): $H_{\text{UQFF}} = 0.987$ vs flat-universe $H_{\text{UQFF}0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0 \Rightarrow H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^{-8} \text{ kg/m}$.

Numerical value for Andromeda:

$$\begin{aligned} a_{\text{dust}} &= \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\ &= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\ &= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2 \end{aligned}$$

At ~ 9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. $M_{\text{visible}} / M_{\{\text{DM}_{\text{mass}}\}}$ Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$\begin{aligned} M_{\text{visible}} &= (1 - f_{\text{DM}}) \times M \\ M_{\text{DM, mass}} &= f_{\text{DM}} \times M \end{aligned}$$

For Andromeda defaults: $M = 1.989 \times 10^4 \text{ kg}$, $f_{\text{DM}} = 0.80$: - $M_{\text{visible}} = 0.20 \times 1.989 \times 10^4 = \mathbf{3.978 \times 10^4 \text{ kg}}$ (20% visible baryons) - $M_{\{\text{DM}_{\text{mass}}\}} = 0.80 \times 1.989 \times 10^4 = \mathbf{1.591 \times 10^4 \text{ kg}}$ (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 2632), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA_{UQFF_MODULE}.cpp g_{total}

The full UQFF 2.0 equation for Andromeda with all PAPER_273276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g, \text{sum}}^{(26)} + \Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at $t=0$ (current epoch):

Term	Value (m/s)	Paper
$g_{\text{grav}} = \mu_s \nabla(M_s/r)$	$1.227 \times 10^?$	baseline
$U_{g, \text{sum}}$ (26 layers)	$6.380 \times 10^{??}$	26-layer Triadic
Λ -term	$2.998 \times 10^?6$	cosmological

Term	Value (m/s)	Paper
<code>g_quantum</code>	$\sim 2 \times 10^6$	HUP
<code>g_Lorentz</code>	$\sim 2 \times 10^6$	IGM EM
<code>g_fluid</code>	$\sim 8 \times 10^?$	IGM buoyancy
<code>F_res(?_HI) at t=0</code>	$1 \times 10^?$	PAPER_274
<code>g_DM</code>	$1.356 \times 10^?$	PAPER_275
<code>g_expansion at t=0</code>	0	PAPER_276
<code>a_dust</code>	$4.29 \times 10^{??}$	PAPER_276
<code>?_approach</code>	1.001001	PAPER_273

Note: `g_expansion` grows linearly with `t`, dominating over `g_grav` at `t = t_H`.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`: 1. `H0 = 70.0 km/s/Mpc` 2. `Omega_m = 0.3` 3. `Omega_Lam = 0.7` 4. `Mpc_{to\_m} = 3.086 \times 10^?` 5. `rho_dust = 1 \times 10^? kg/m` 6. `M_visible = (1f_DM)M` 7. `M_{DM\_mass} = f_DMM`

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

1. **Friedmann-UQFF Expansion Coupling:** `g_expansion = g_base - H(z) t`. At `t = t_H`, adds 98.7% of `g_base` near-gravitational doubling over cosmological time.
2. **H_UQFF = 0.987:** New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 t_H \approx 1$ in flat Λ CDM. Andromeda's blueshift ($z = -0.001$) makes `H_UQFF` 0.15% below flat-universe value a blueshift Friedmann suppression.
3. **M_visible/M_{DM_mass} cascade + dust drag:** Explicit DM split bookkeeping (update-Cache) and ISM dust ram-pressure (`a_dust` $4 \times 10^{??}$ m/s).

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`.Groups[1].Value`

`H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as $g_{\text{expansion}} = (\mu s \nabla(M_s/r)) H(z) t$. For Andromeda's blueshift redshift $z = -0.001$, the Friedmann $H(z)$ equals $69.969 \text{ km/s/Mpc} = 2.269 \times 10^? s^{-1}$. Evaluated at the Hubble timescale $t_H = 4.352 \times 10^{-7} s$, the dimensionless Friedmann-UQFF Resonance Coefficient $H_UQFF = H(z) \cdot t_H$ **0.987** a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the `M_visible/M_{DM\_mass}` cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

1. Background

The UQFF framework computes total gravitational acceleration $g_{\text{total}}(r,t)$ as a sum of DPM-seeded, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER_273275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter $H(z)$. This is significant because over a Hubble timescale, $H(z)t \approx 1$ implying that expansion-mediated gravitational coupling is of the same order as the Step 10 Newton observational projection term.

2. Friedmann $H(z)$ Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard Λ CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

For Andromeda parameters: $H_0 = 70.0$ km/s/Mpc, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $z = -0.001$.

$$\begin{aligned} H(z = -0.001) &= 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7} \\ &= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991} \\ &= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc} \end{aligned}$$

In SI units: $H_{\text{SI}} = 69.969 \times 10^{-3} / 3.086 \times 10^7 = 2.269 \times 10^{-8} \text{ s}^{-1}$

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t .

Numerical values for Andromeda: - $g_{\text{base}} = \mu_s \nabla(M_s/r) = 6.674 \times 10^{-11} \cdot 1.989 \times 10^{30} / (1.04 \times 10^7)^2 = 1.227 \times 10^{-7} \text{ m/s}^2$ - $Att = 1 \text{ Gyr} (3.156 \times 10^8 \text{ s}) : g_{\text{expansion}} = 1.227 \times 10^{-7} \cdot 2.269 \times 10^{-8} \times 3.156 \times 10^8 = 8.79 \times 10^{-7} \text{ m/s}^2$ - $Att = t_{\text{Hubble}} (4.352 \times 10^8 \text{ s}) : g_{\text{expansion}} = 1.227 \times 10^{-7} \approx 0.987 = 1.211 \times 10^{-7} \text{ m/s}^2$

3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^{-7}$ s (Hubble time 13.8 Gyr).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

$H_{\text{UQFF}} \approx 0.987$ is a remarkable result. It means:

Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the Step 10 Newton observational projection term** nearly doubling g_{base} .

This near-unity value is not coincidental. In a flat Λ CDM universe ($O_m + O_\Lambda = 1$):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because $H_0 t_H$ is the dimensionless Hubble number ≈ 0.96 for observationally consistent H_0 .

New UQFF Constant: $H_{\text{UQFF}} = H(z) \cdot t_H$ **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H_0	70.0 km/s/Mpc
O_m	0.3
O_Λ	0.7
z (Andromeda)	-0.001
$H(z)$	$2.269 \times 10^{-18} \times t_H \times 4.352 \times 10^{-7} \times H_{\text{UQFF}} \times 0.987 \times (1 + g_{\text{expansion}}(t_H) \times 1.211 \times 10^{-27} \times g_{\text{base}} \times 1.227 \times 10^{-27})$
Ratio $g_{\text{exp}}/g_{\text{base}}$	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $O_m(1+z)$ is suppressed below O_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z = 0)$$

For Andromeda ($z = -0.001$): $H_{\text{UQFF}} = 0.987$ vs flat-universe $H_{\text{UQFF}0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0 \Rightarrow H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^{-8} \text{ kg/m}$.

Numerical value for Andromeda:

$$\begin{aligned} a_{\text{dust}} &= \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\ &= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10} \\ &= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2 \end{aligned}$$

At ~ 9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. $M_{\text{visible}} / M_{\{\text{DM}_{\text{mass}}\}}$ Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$\begin{aligned} M_{\text{visible}} &= (1 - f_{\text{DM}}) \times M \\ M_{\text{DM, mass}} &= f_{\text{DM}} \times M \end{aligned}$$

For Andromeda defaults: $M = 1.989 \times 10^4 \text{ kg}$, $f_{\text{DM}} = 0.80$: - $M_{\text{visible}} = 0.20 \times 1.989 \times 10^4 = 3.978 \times 10^4 \text{ kg}$ (20% visible baryons) - $M_{\{\text{DM}_{\text{mass}}\}} = 0.80 \times 1.989 \times 10^4 = 1.591 \times 10^4 \text{ kg}$ (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 2632), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA_{UQFF_MODULE}.cpp g_{total}

The full UQFF 2.0 equation for Andromeda with all PAPER_273276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g, \text{sum}}^{(26)} + \Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at $t=0$ (current epoch):

Term	Value (m/s)	Paper
$g_{\text{grav}} = \mu_s \nabla(M_s/r)$	$1.227 \times 10^{-26} \text{ m/s}^2$	PAPER_273
$g_{\text{expansion}} = H(z) t$	$2.998 \times 10^{-26} \text{ m/s}^2$	PAPER_274
$g_{\text{HUP}} = g_{\text{Lorentz}} $	$1.356 \times 10^{-26} \text{ m/s}^2$	PAPER_275
$g_{\text{dust}} = a_{\text{dust}} g_{\text{expansion}} $	$4.29 \times 10^{-26} \text{ m/s}^2$	PAPER_276
g_{approach}	1.001001	PAPER_273

Note: $g_{\text{expansion}}$ grows linearly with t , dominating over g_{grav} at $t = t_H$.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`: 1. $H_0 = 70.0 \text{ km/s/Mpc}$ 2. $\Omega_m = 0.3$ 3. $\Omega_{\text{Lam}} = 0.7$ 4. $\text{Mpc}_{\text{to_m}} = 3.086 \times 10^5 \text{ m}$ 5. $\rho_{\text{dust}} = 1 \times 10^{-26} \text{ kg/m}^3$ 6. $M_{\text{visible}} = (1f_{\text{DM}})M$ 7. $M_{\text{DM_mass}} = f_{\text{DM}} M$

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

1. **Friedmann-UQFF Expansion Coupling:** $g_{\text{expansion}} = g_{\text{base}} - H(z) t$. At $t = t_H$, adds 98.7% of g_{base} near-gravitational doubling over cosmological time.
2. **$H_{\text{UQFF}} = 0.987$:** New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 t_H = 1$ in flat Λ CDM. Andromeda's blueshift ($z = -0.001$) makes H_{UQFF} 0.15% below flat-universe value a blueshift Friedmann suppression.
3. **$M_{\text{visible}}/M_{\text{DM_mass}}$ cascade + dust drag:** Explicit DM split bookkeeping (update-Cache) and ISM dust ram-pressure ($a_{\text{dust}} = 4 \times 10^{-26} \text{ m/s}^2$).

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UQFF computed: MUGE buoyancy ratio $U_{\text{bi}}/F_{\text{U}} = [SSq] r/GM = 5.7e-15 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7 \text{ m/s}^2$ at r_{ISCO} .

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\text{U_Bi_i}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M– σ correction (PAPER_1048): The phonon-corrected M– σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: SCm-Modified NFW Dark Matter Profile

Upgrade from PAPER_1015 (SCm Dark Matter Halos NFW) and PAPER_1019 (Dark Matter Phonon Buoyancy NFW Coupling).

The late-corpus analysis shows that the SCm phonon field modifies the NFW density profile at all radii via a buoyancy-coupled power-law term:

$$\rho_{\text{UQFF}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2} \times \left[1 + H_{\text{SCm}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{r_s}{r}\right)^{\alpha_{\text{phonon}}} \right]$$

where: - $\alpha_{\text{phonon}} = 0.3$ governs the radial decay of phonon coupling - $\beta_i = 0.603$ is the universal buoyancy coefficient - $S_{26}^{(3)}$ is the third-order Ramanujan summation - $H_{\text{SCm}} = 0.99$ is the manifold completeness factor

Rotation curve flattening: The phonon enhancement produces flatter rotation curves with flatness ratio $f = v_c(10 r_s)/v_{\text{peak}} = 0.891$, compared to pure NFW $f \approx 0.75$. Peak circular velocity $v_{\text{peak}} \approx 204 \text{ km/s}$ for $M_{\text{halo}} = 10^{12} M_{\odot}$, $c = 10$.

Halo stabilization: The effective buoyancy pressure $P_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot \beta_i$ prevents cusp-core divergence, providing a physical mechanism for observed cored profiles without invoking SIDM cross-sections.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.064 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 17, \quad n_{\text{channel}} = 17/26$$

Since $p_{\text{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m / M_{\odot}} \right) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH},\text{sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.064	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 17$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} m^{-2}$	Planck 2018	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p suppression}}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{\text{U_Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\text{U_Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\text{U_Bi_i}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\text{U_Bi_i}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback

Paper	Title
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1015	SCm Dark Matter Halos NFW Rotation Curve
PAPER_1019	Dark Matter Phonon Buoyancy NFW Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed

11 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\ramanujan\appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] * n / 26)$
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U, Bi\}} / F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma^2)] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1 - 2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer $(a; q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q; q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2; q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where
 $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{ppai}} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ_{ppai}	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β_{i}	0.603	Buoyancy coupling coefficient
H_{SCm}	0.99	SCm manifold completeness
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
σ_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`,
`MAIN_{1\CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`,
`uqff_{mock\_theta\_pi\_kernel}.wl`).

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